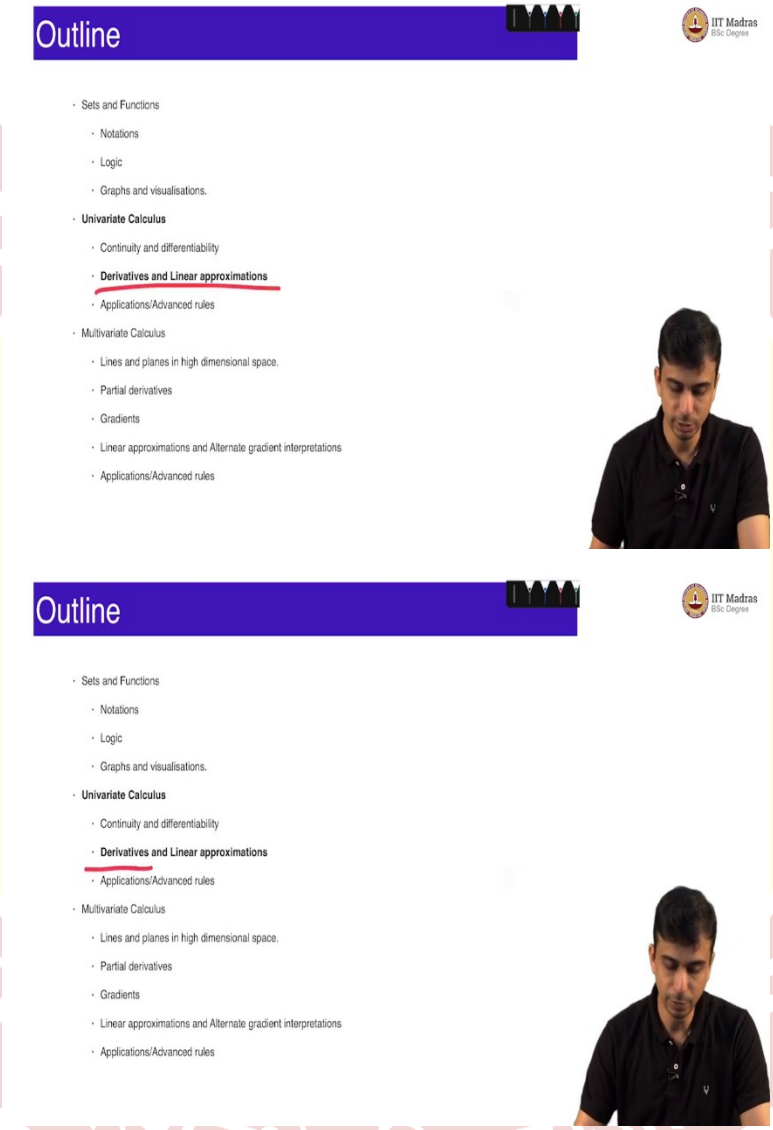


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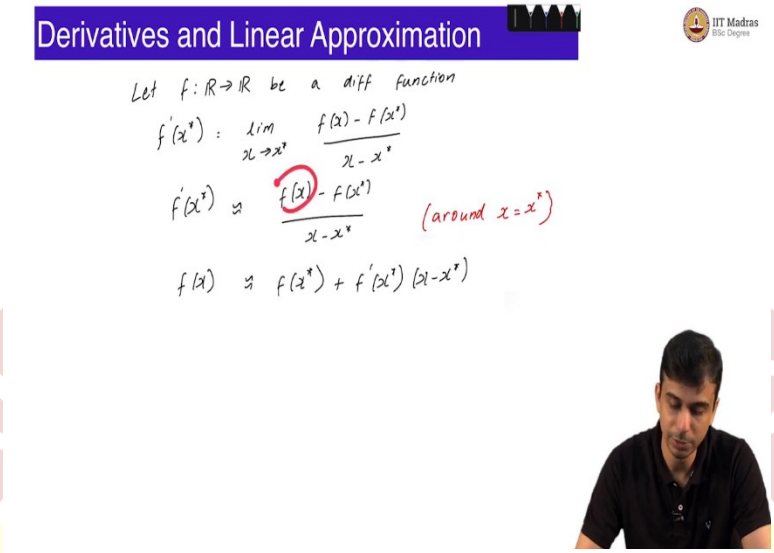
Machine Learning Foundation
Assistant Professor Harish Guruprasad Ramaswamy
Department of Computer Sciences and Engineering
Indian Institute of Technology, Madras
Univariate Calculus: Derivative and Linear Approximations

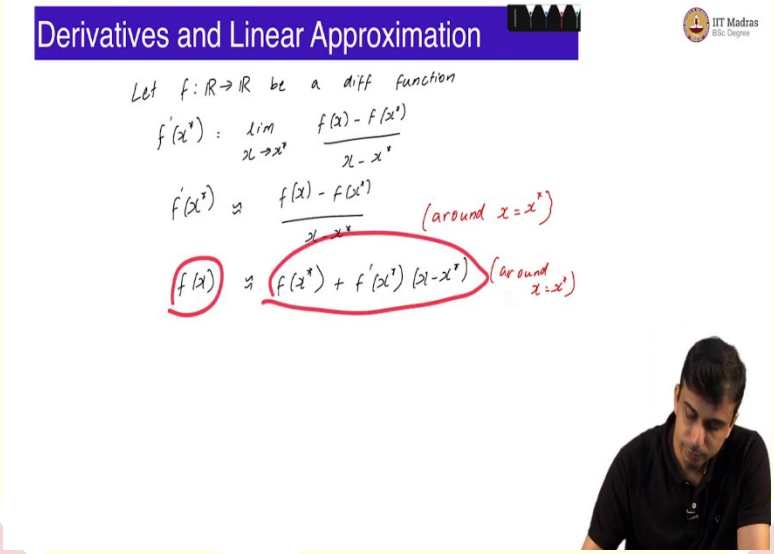
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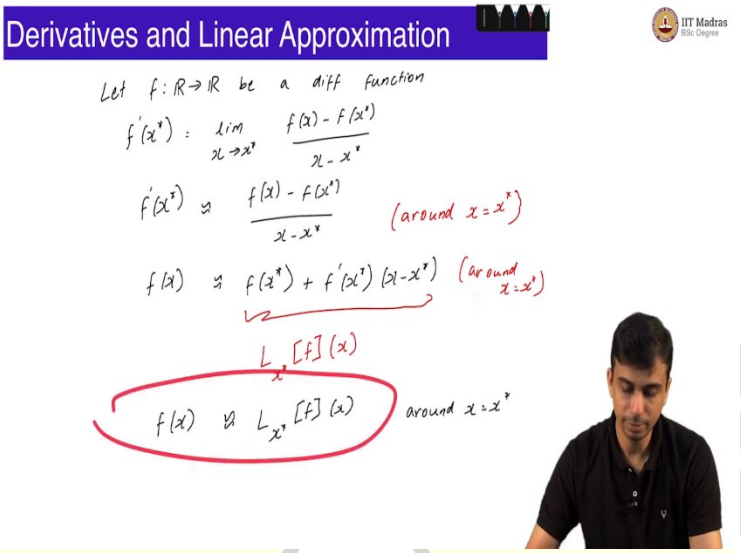


Hello, everyone, and welcome to another lecture of machine learning foundations. We will continue with the, the univariate calculus ideas that we had begun, last class. In this class, we will mainly focus on the derivative, and its link to what are called linear approximations.

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So, let us write down, let us consider a differentiable function from R to R. So, let f from R to R be a differentiable function. What is the definition of the derivative? Well, f dash of x * = limit x approaching x * f of x - f of x * by x - x *. So, we do have a limit term here. So, we cannot do direct multiplication by x - x * and you cannot move this here.

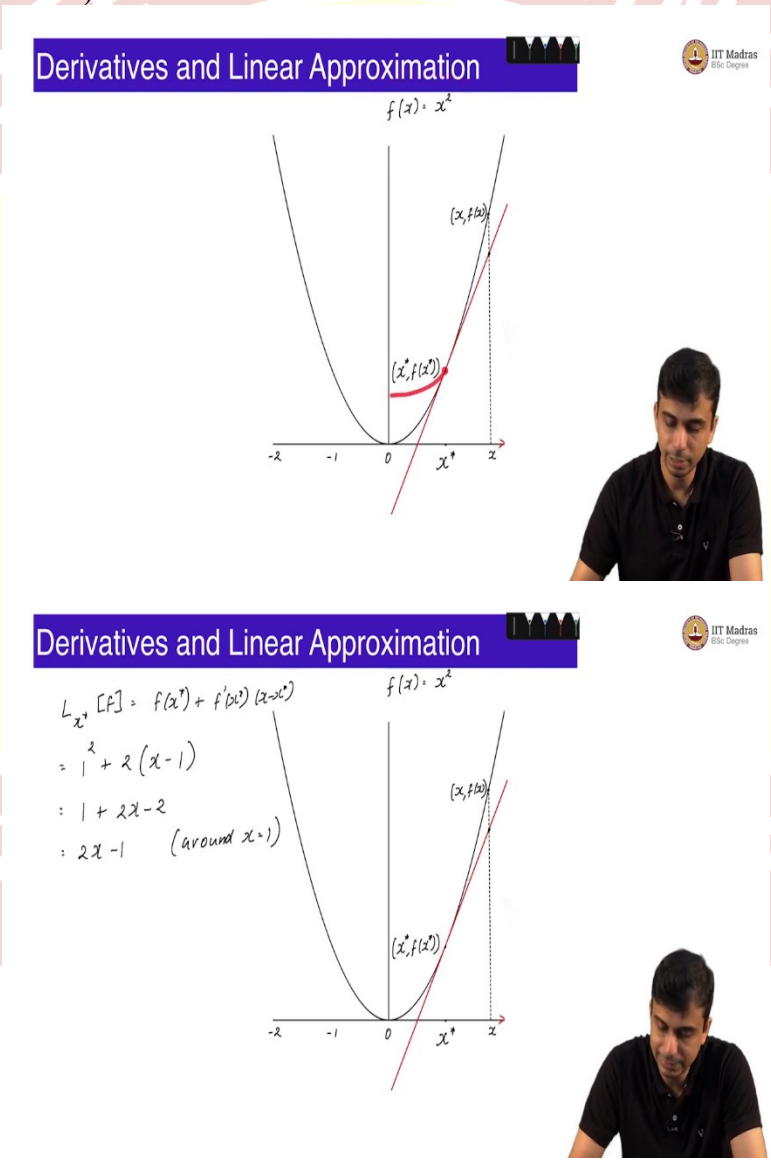
But let us assume that. So, let us assume that x is approximately= x *. That is, we are in a regime where x is approximately equal x *, in which case the limit x equal x approaching x * is satisfied. So, we can say that f of x * is approximately= f of x - f of x * by x - x *, in the neighborhood around x equals x *. So, this is true only when x is approximately= x *.

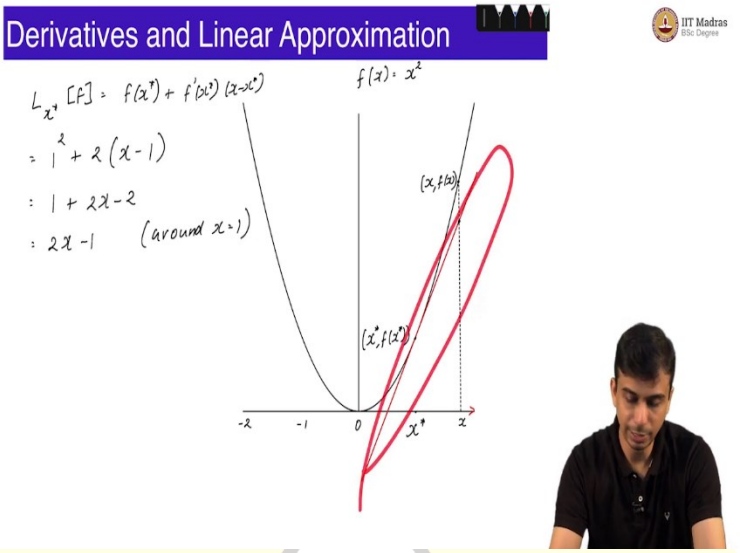
Only then, we can remove this limit statement without affecting logic. Now, that we have this, we can move things around and write an alternative expression, which is f of x is approximately= f of x * + f dash of x * into x - x *. So, f dash of x * is approximately= f of x - f of x * by x - x * around x= x *, which means that f of x is approximately= f of x * f dash of x * into x - x * + f of x *.

This is also true around x = x *. When x is close to x * this is true. And this is called a linear approximation. So, you can view this, the left-hand side is a function of x. The right-hand side also can be viewed as a function of x, you can view x * and f dash x * as constants, which means that f of x * is a constant, f dash of x * is also constant. This entire term also can be viewed as a function of x.

And we are going to denote that by L x * of f. This entire thing is a function of f. And if you want to evaluate it at a point x, so, it is L x * of f of x. So, what the approximately= statement says is f of x is approximately= L x * of f evaluated at x around x = x *. This is the key idea, the key expression in essentially, first order calculus. So, this is f of x can be approximated by f of x * + f dash f x * index - x * around x equals x * is the, is the key approximation idea that we will be using for a major part of machine learning.

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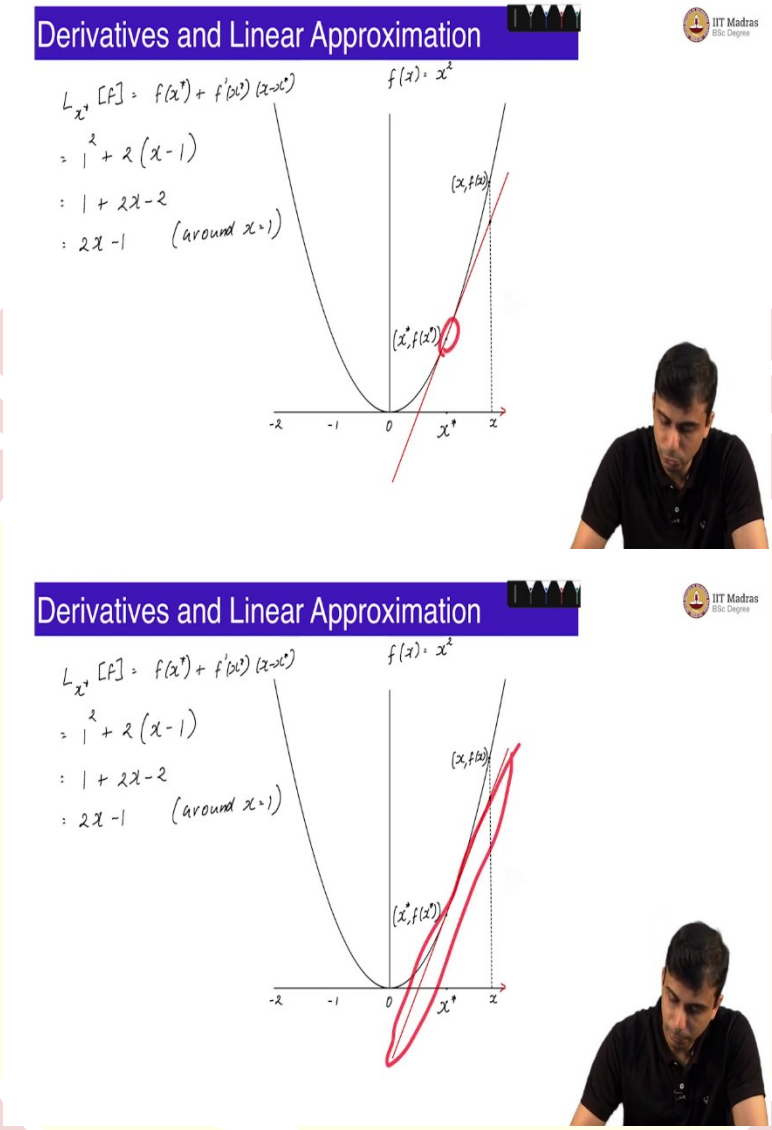


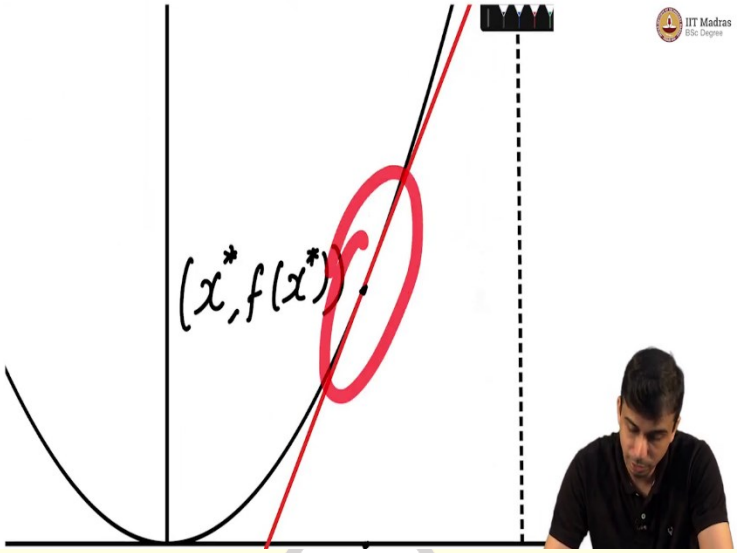


So, what does this mean for us? So, let us give some graphical illustration of this. Let us consider the simple function f of x= x squared, which we know to be differentiable. And let us pick some x * is equal r= 1. So, let us pick x * = 1, and consider to the point x * , f of x *. What is f dash of x *? F dash of x * is simply= 2, that simply follows from calculus.

So, f of x is x squared f dash of x is 2x, f dash of x * is 2. So, now we can write down what the linear approximation actually is. What is a linear approximation? Well, L x * of f is simply= f of x * + f dash of x * into x - x *, which happens to be 2 squared + 2 into x - 2, which is 4 + sorry, it is 1 squared + 2 into x - 1. We said x * = 1, so, 1 + 1 squared + 2 into x - 1, which is 1 + 2x - 2, which = 2x - 1. So, what we are saying is the linear approximation of f around x, around 1, around x= 1, this is valid around x= 1. This red line corresponds to the graph of 2x - 1.

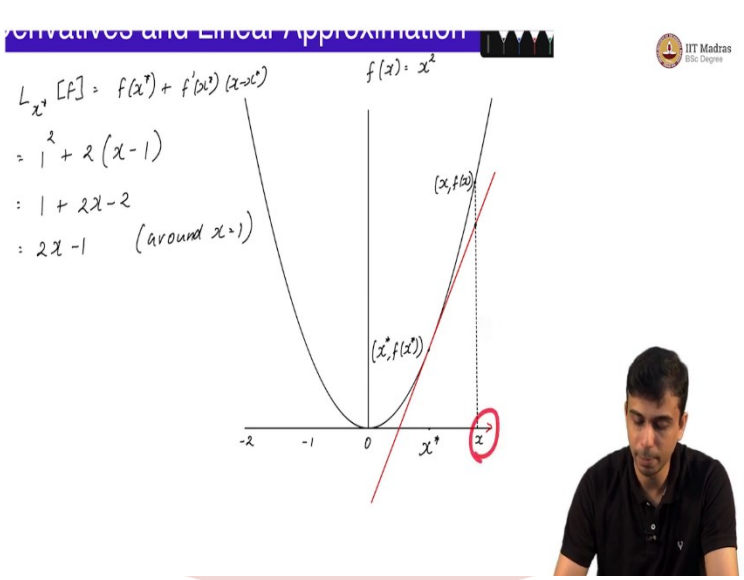
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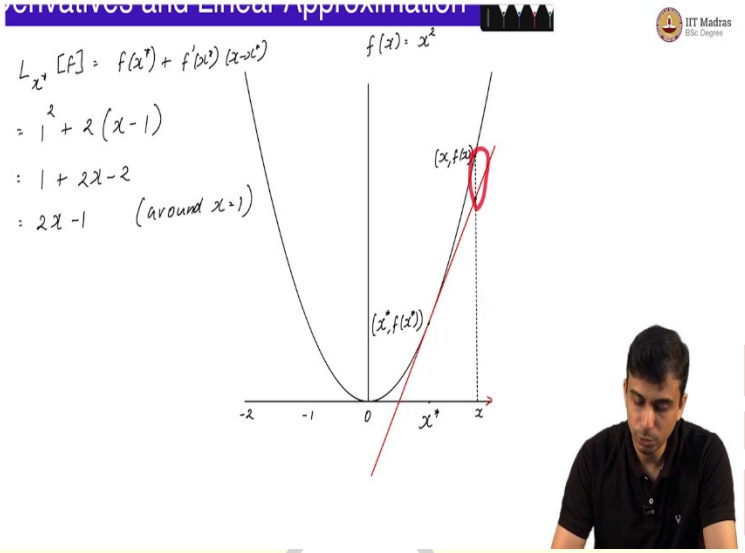


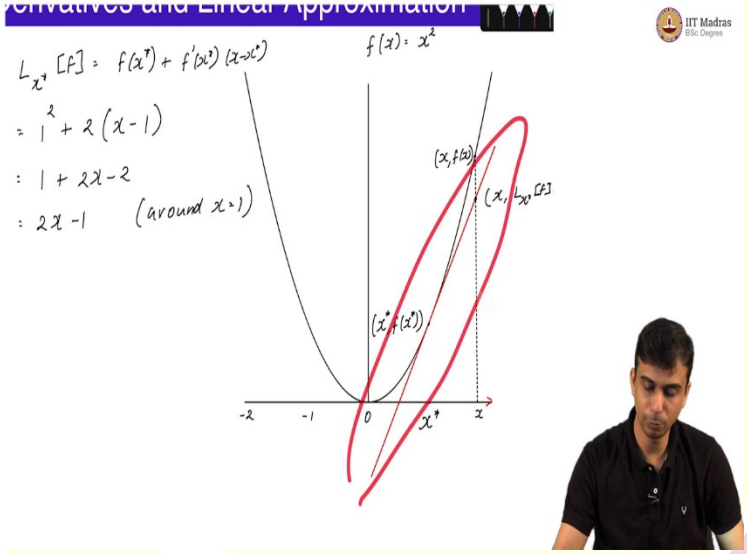


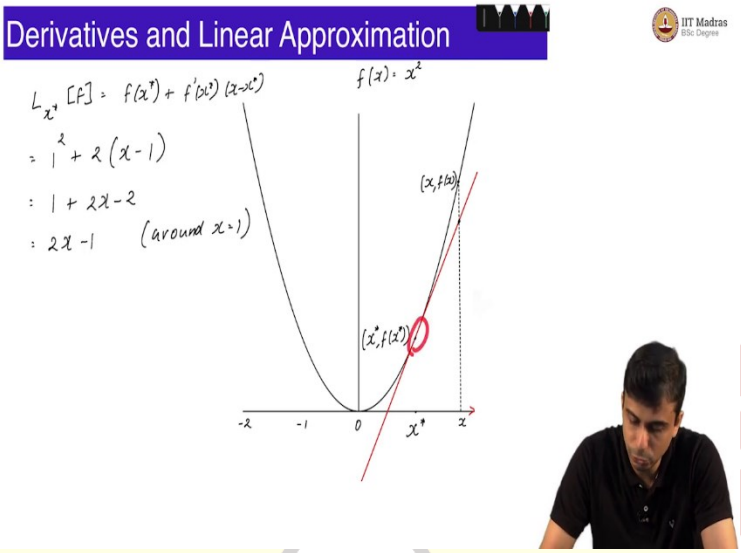
And you can see that this particular graph is an approximation to the black curve, which corresponds to x squared around x * = 1. So, you can see that x * = 1, it exactly touches it, it is exact. The farther your x moves from x *, the worse your approximation is, around x = x *, your red line and black line coincide quite well.

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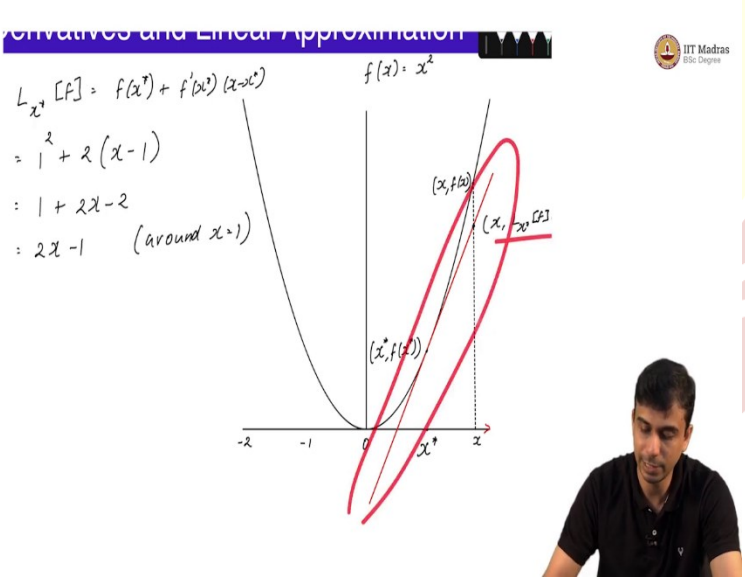

Derivatives and Linear Approximation
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$L_{x^*}[F] = f(x^*) + f'(x^*)(x-x^*)$
$= 1^2 + 2(x-1)$
$: 1 + 2x - 2$
$: 2x - 1$ (around $x=1$)
$f(x) = x^2$
$(x, f(x))$
$(x^*, f(x^*))$
-2
-1
0
x^*
x

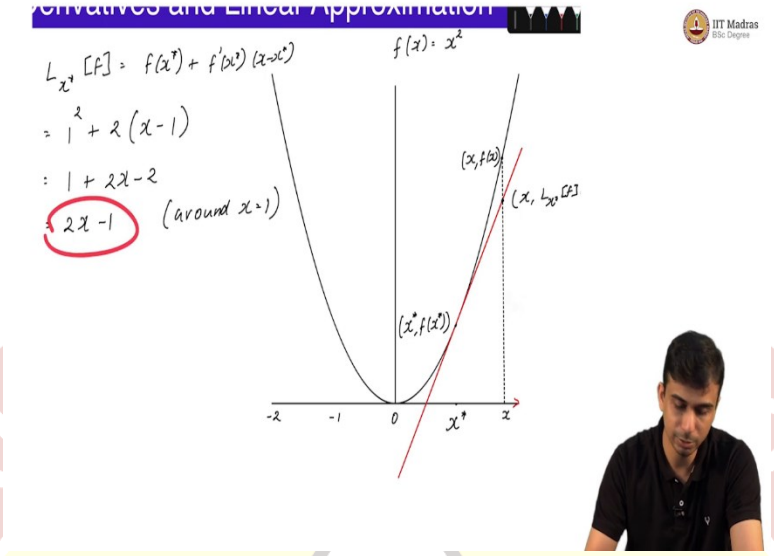

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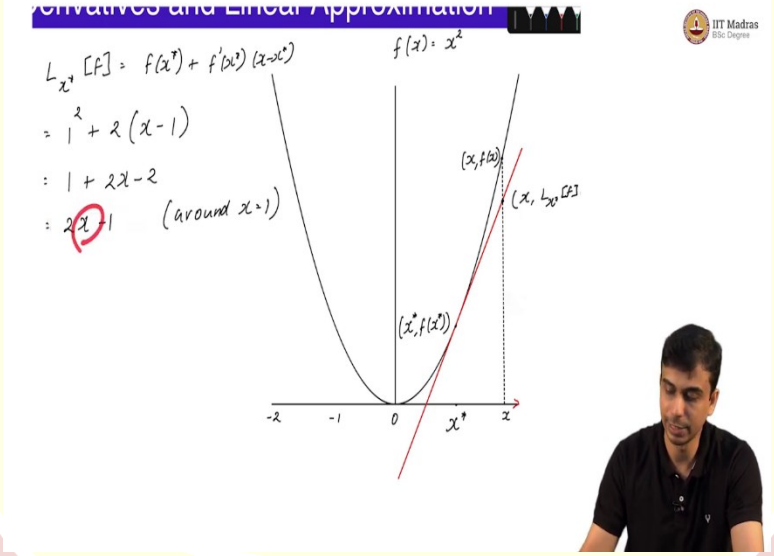


That is the reason why it is a linear, it is a approximation. Assume, as your x moves farther and farther away from x *, it becomes a (())(07:17) approximation. For example, at this point, let us say x = 2, we can see that there is actually a gap between f of x and L x * of f evaluated at x. So, this black, this point here corresponds to x , L x * of f evaluated at x. So, that is the reason why I am saying that L x * of f is an approximation of f around x= x *.

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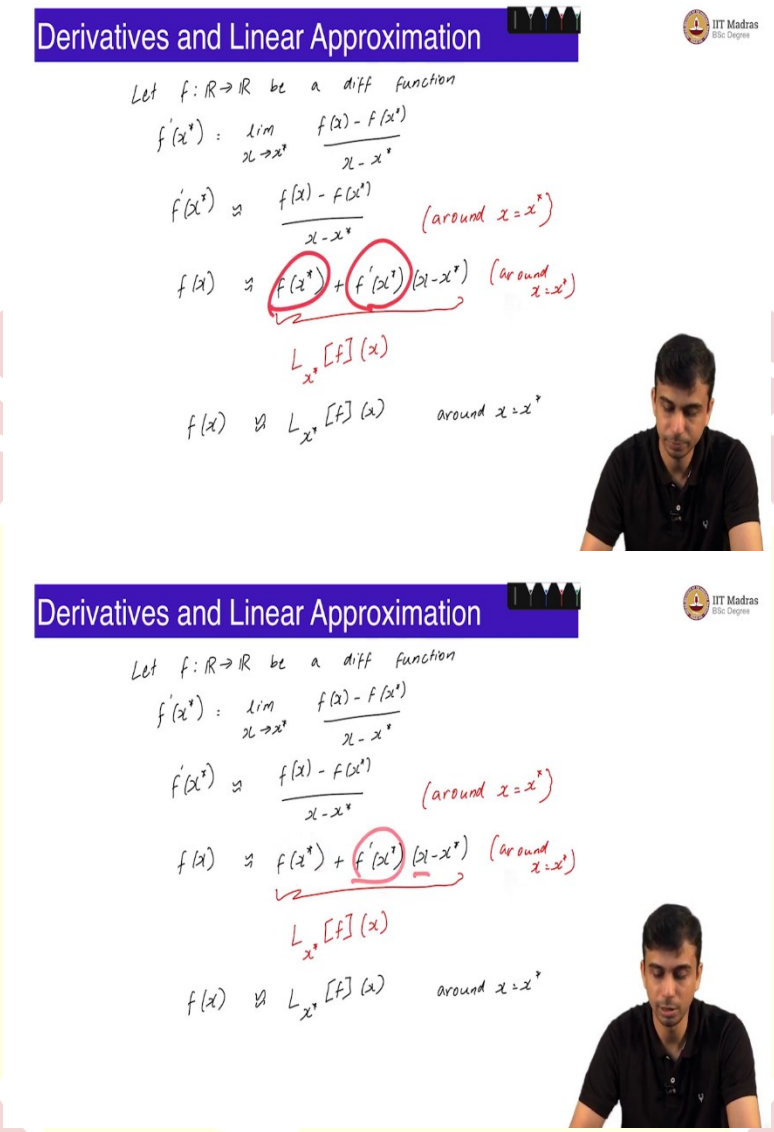


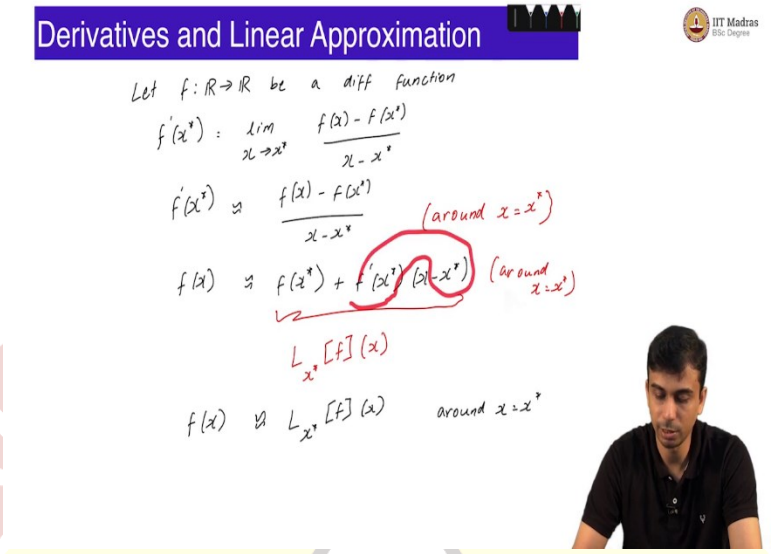


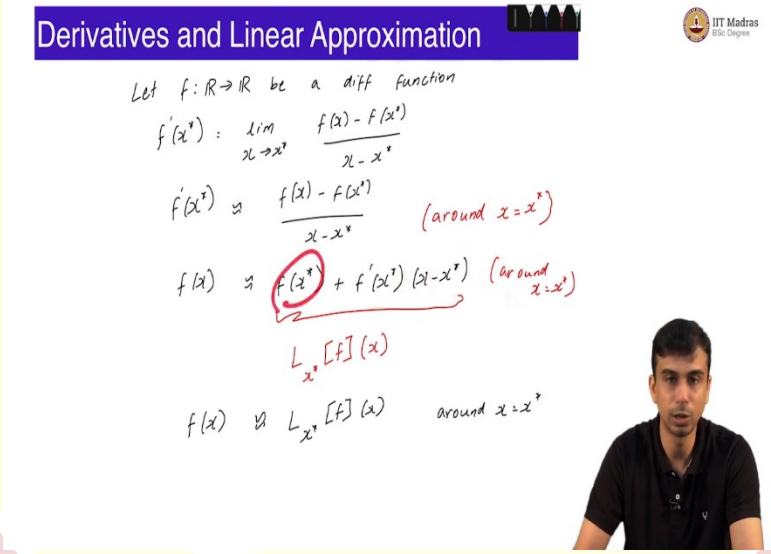


So, that is, that is what we mean by saying that L x * of f is a linear approximation. And why is it a linear approximation? You can see that L x * of f, the graph of Lx * of f is a straight line. Why is that? Because this 2x - 1 is a linear function of x, the coefficient is it is, it is a equation of a straight line.

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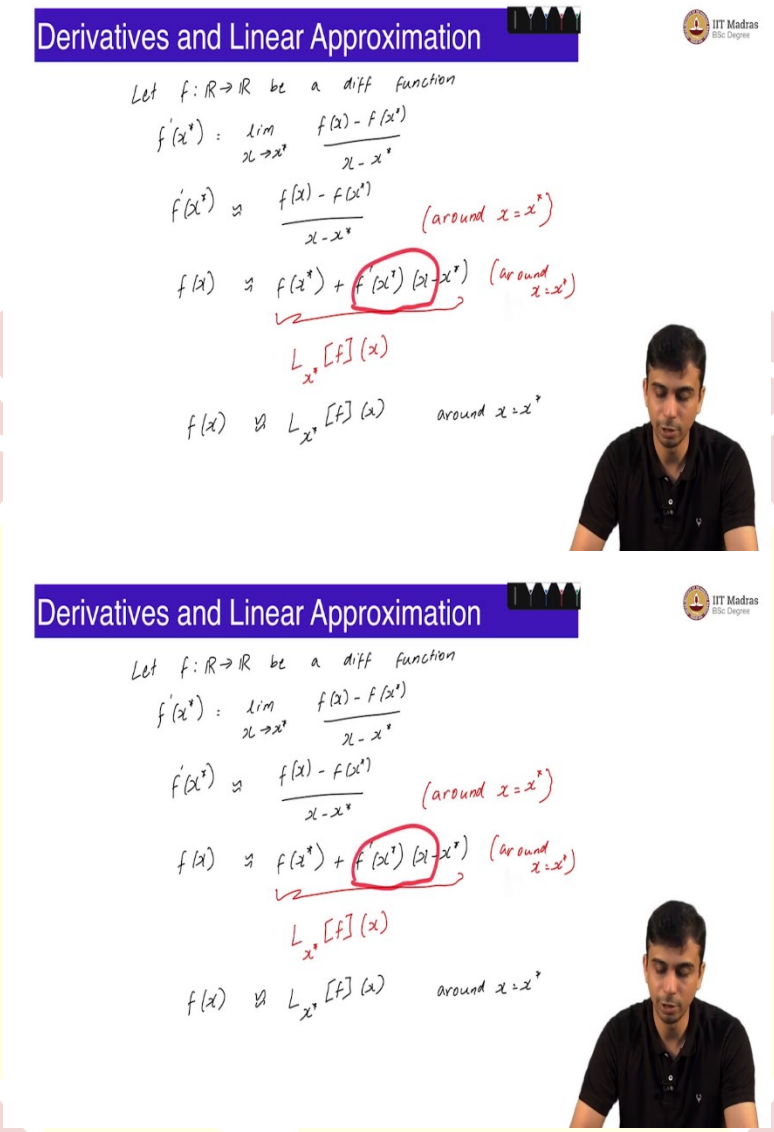


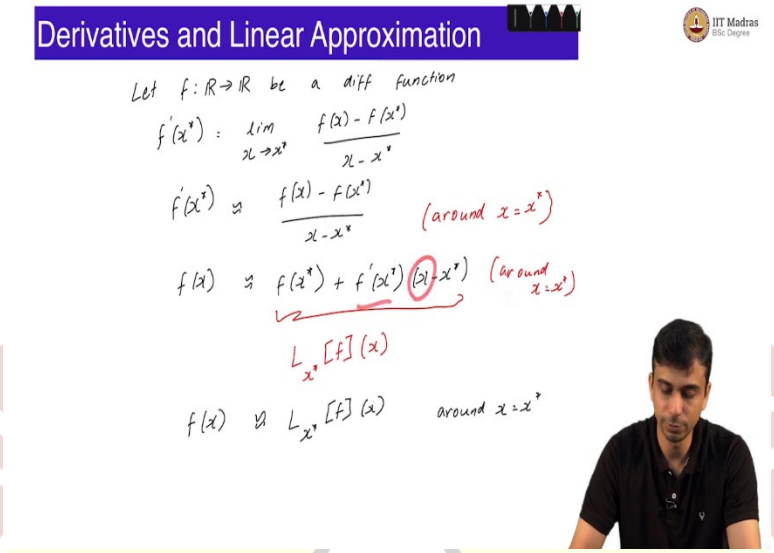


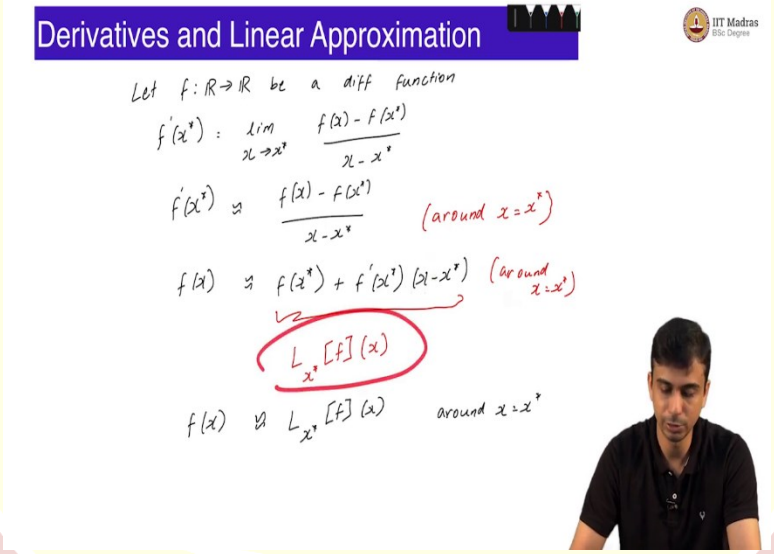


In general, also, it is true because you can see that L x * of f viewed as a function of x is simply the coefficient is f dash x * which is a constant. We have, this as a constant and constant times x, and f dash x * into - x * is also a constant.

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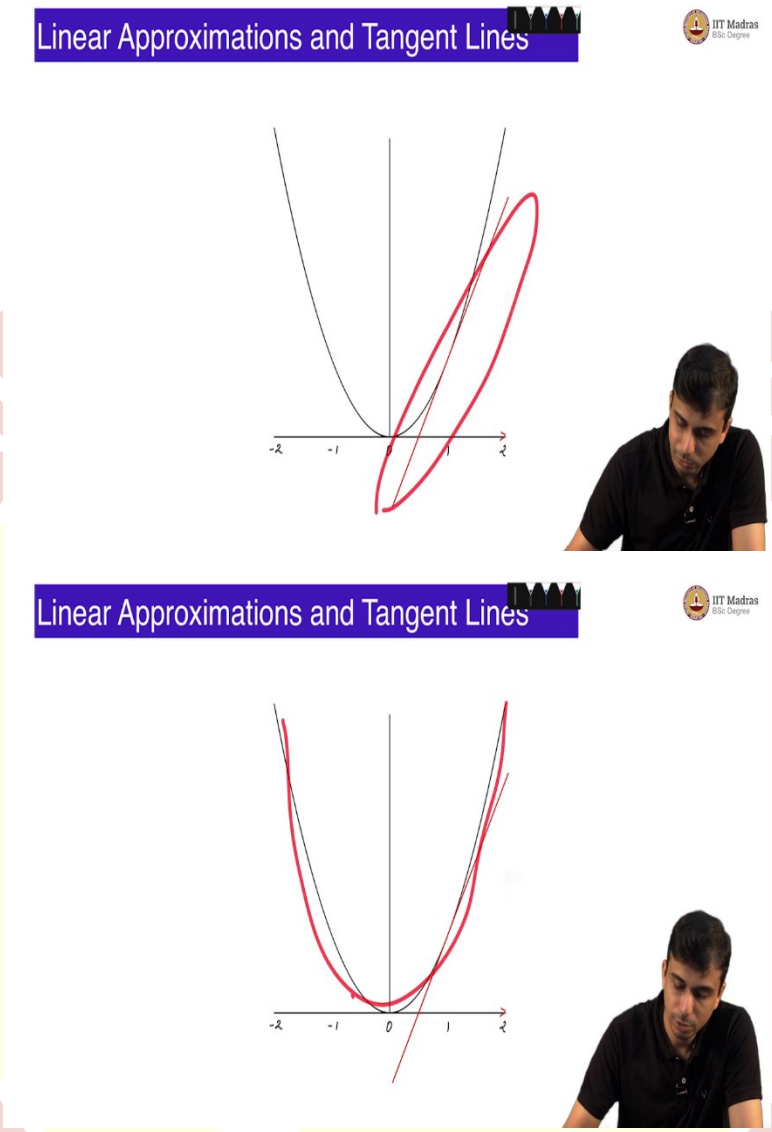


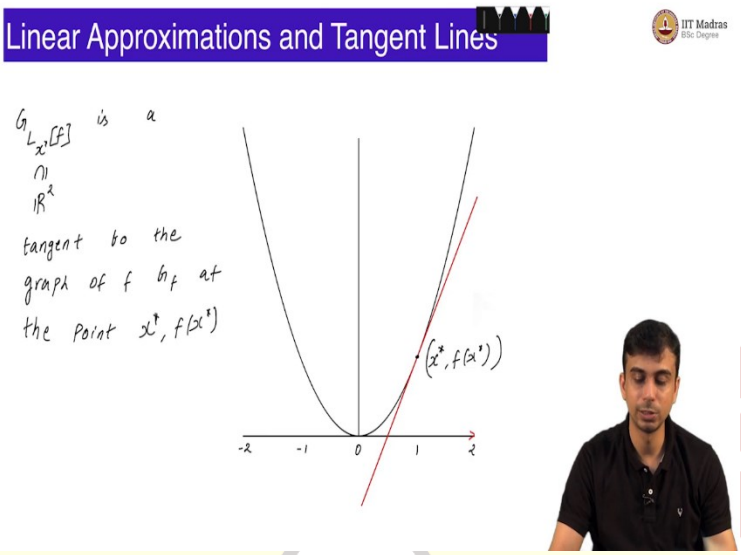




So, f of x * + f dash of x *, sorry, f of x * - f dash x * expression is a constant. And you have f dash x * into x, which is a linear coefficient of x multiplied by x. So, that is the reason why this is called a linear function. And this is an approximation of f around x= x *.

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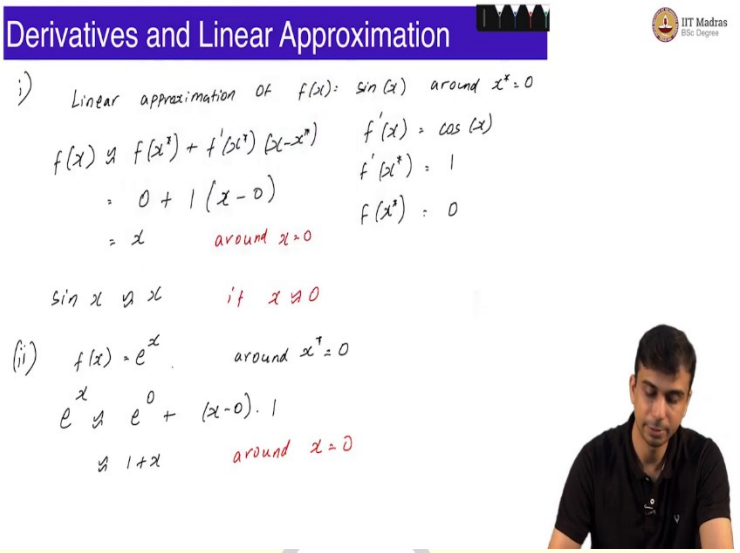


So, that is, that is about linear approximations. And you can, you would also seen another interpretation of linear approximation via what is called tangent lines. So, tangent is a, a geometric object, while a linear approximation is, a functional object. What is a tangent? Well, a tangent is, I will geometrically define vaguely dominantly define, one set is a tangent to another set if it touches the other set.

So, if to, we can say a line is a tangent to a set, if that line touches the first set. I am not going to define touching in a formal sense because that is way too much work, but it is. In this case, you can see that the red line which corresponds to the graph of the linear approximation of f around x * is a tangent to the graph of f. Or in other words, I am going to write it in a formal way, which is your graph of L x * of f, L x * of f is also a function.

So, the graph of that function is also going to be a subset of R2, in the case of functions from R to R, this is a subset of R2, is a tangent to the graph of f Gf at the point x * , f of x *. So, this point is the tangent, is the point at which these two sets touch each other. That is x * , f of x *. This is the definition of a tangent line, a linear approximation to form tangent lines to the graph, the graph of linear approximations forms tangent lines to the graph of the function.

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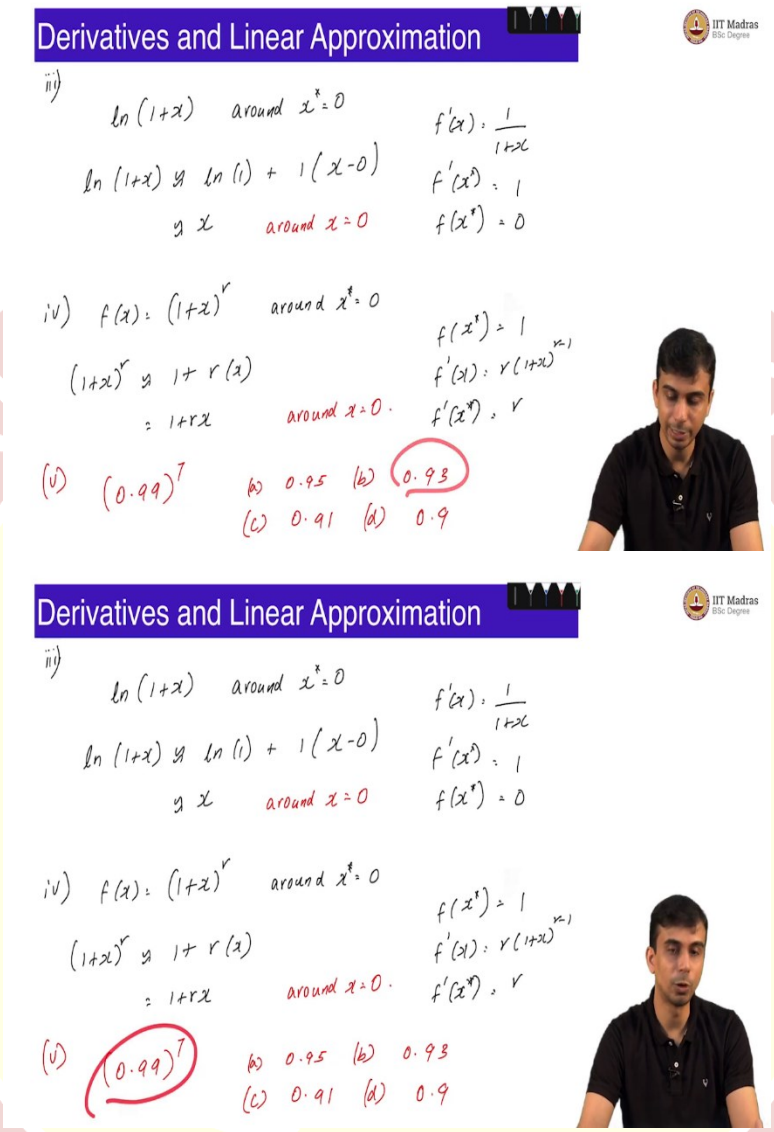


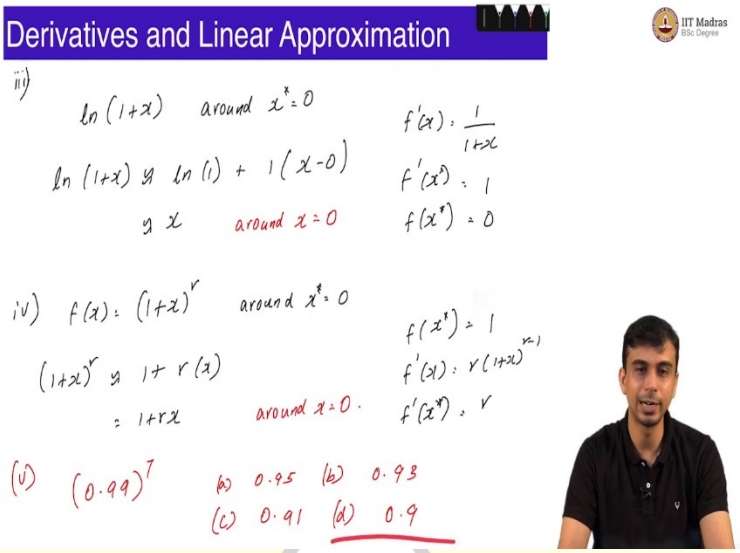
Here. now we can, we have the basics of linear approximations and tangent lines, we can see some examples. The first examples perhaps the most famous one, linear approximation, give the linear approximation of f of x = sin x around x * which = 0. So, to do that, you need f dash of x. What is f dash f of x? F dash of x is simply cos x, f dash of x * just (())(12:08) 0, which is 1. And f of x *, which is sin 0, which is 0.

With these three facts put together, we can say f of x is approximately= f of x * + f dash of x * times x - x *. What is that? Which is 0 + 1 into x - 0, which is and when is this valid? This is valid around, around x = 0. When x is approximately, x is close to 0. This is a value approximation, or in other words, it says that sin x is approximately= x, if x is approximately= 0.

This is the famous sin theta by limit, sin theta by theta = 1 thing that you would have derived in your school. Let us do one more classic (())(13:18). Let us say f of x = e power x. I want the linear approximation of this function around x * = say, 0. In this case, what would be, what would be e power x? That is approximately= e power 0, + x - 0 into f dash of 0, which = 1. So, e power x is approximately= 1 + x around x = 0. So, I am going to highlight the fact that e power x is approximately 1 + x is true only around x= 0. And I am going to highlight that, similarly this also deserves a highlight.

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So, a few more classic examples. Let us do what is log of 1 + x around x * = 0. F of x= log 1 + x, what is f dash of x? F of x= 1 by 1 + x, f dash of x * = 1. And f of x * = 0. So, we can put all that together and say log of 1 + x is approximately= log of 1, which is 0 + f dash of x *, which is 1 into x - 0. So, log of 1 + x is approximately= x around x = 0.

So, we will do one more example. Let us say f of x = 1 + x power r, r is some integer, let us say 2, or 3 or 7 it does not matter. So, 1 + x power r around, you want approximate 1 + x power r, around x = 0. So, what is f of, what is f of x *? F of x * = 1, f dash of x in general is simply= r into 1 + x, or r - 1. And f dash of x * in particular, is going to be= r.

We can put all of that together and say that 1 + x power r is approximately= 1 + r into x - 0, (())(16:21) just 1 + r x. And this is also valid around x= 0. So, this should give you a idea of all, pretty much a lot of calculus makes approximations very easy to write down. So, this is e power x is approximately 1 + x around x = 0 or log 1 + x is approximately= x around x= 0, and so on.

You can generally apply this to any function that you have, if you want an approximation around, which is valid around a particular point x *, you can just use f of x * + f dash x * into x - x *. And the advantage of the linear approximation is that it is a much more simpler object than the original function. And this is going to be the key idea that is going to drive most of optimization.

Let us apply this particular idea for a simple question. Now, here is a, an exercise question. Which of the four options now uses (())(17:35)? I want 0.99 power 7, which of the four are closest to 0.99

power 7. The four options are, a let us say 0.95, b is 0.93, c is 0.91 and d is 0.9. So, you have four options here, and the exercise is to find which of these four options is closest to 0.99 power 7.

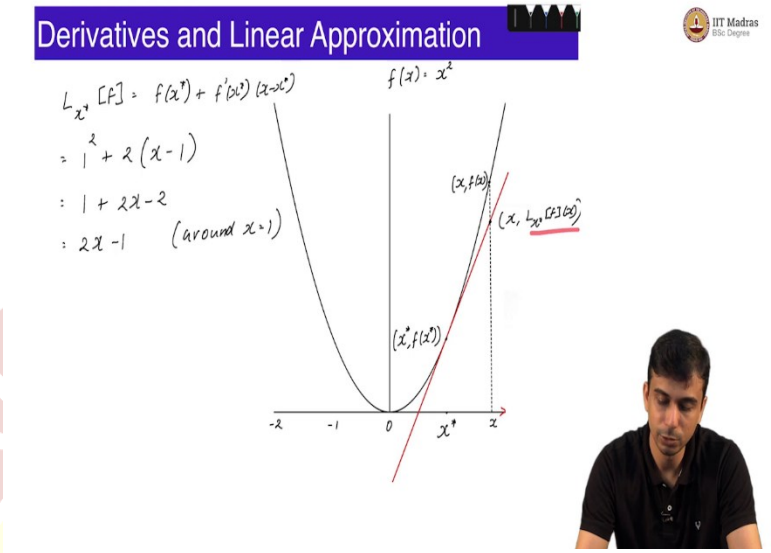
Neither of these are exact, but the question is, which of these is closest. So, 0.99 power 7 should immediately think of 1 + x power 7, where x = - 0.01. And - 0.01 is close enough to 0 that you can apply a linear approximation, which means that the 0.99 power 7 is very easily approximated by 1 + 7 into - 0.01, which is 0.93. So, 0.93 is a (())(18:42) approximation.

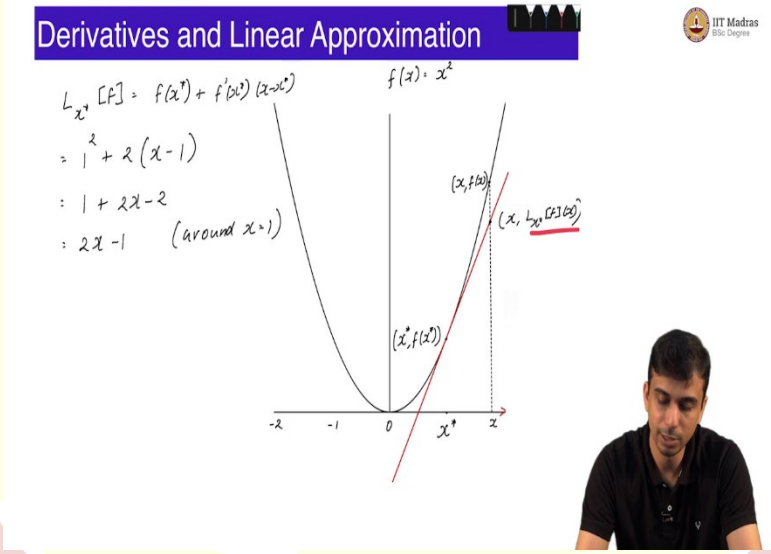
So, you can immediately get an answer of 0.93, for a complex thing like 0.99 power 7, based on the approximations. And intuitively, it makes sense. So, 0.99 power 7 essentially corresponds to a 1 percentage decay of let us say, your money in a bank every year. What (())(19:01) after seven years? How much would you have, how much loss would you have made or how much money would remain in the bank?

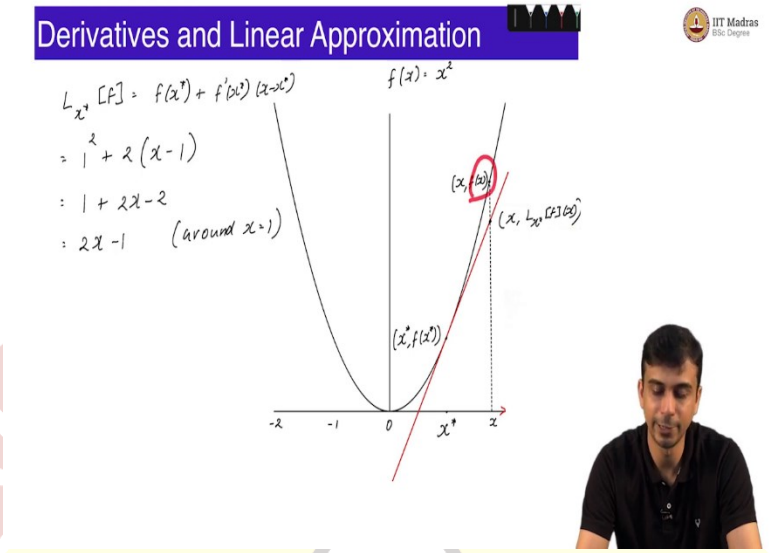
A first approximation would say, well, there is compound interest happening, but it is seven years. So, small enough number of years and one percentage is a small percentage that I can say that 0.99, that is, after 7 years, I would have lost seven percentage, which means that I will be left with 0.93 of my original money.

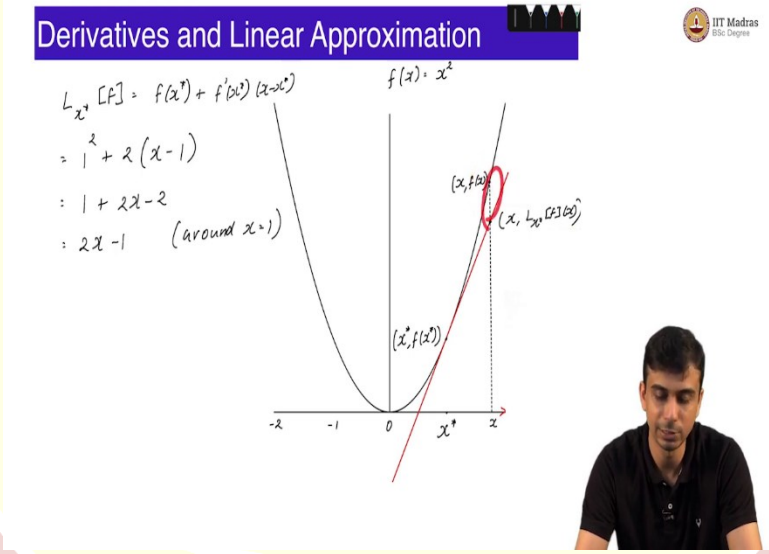
Linear approximation essentially makes this a more formal way. I mean, what approximation are you, by approximating compound interest by simple interest, what you are doing is essentially linear approximations. And this is a valid approximation as long as the rate of interest is small enough, that is, in this case, 1 - 1 percentage is a small enough rate that you can do this in a valid way. And later now, later, we will see that this is not always valid. And in fact, for large enough let us say instead of 1 percentage loss, we had 10 percentage loss.

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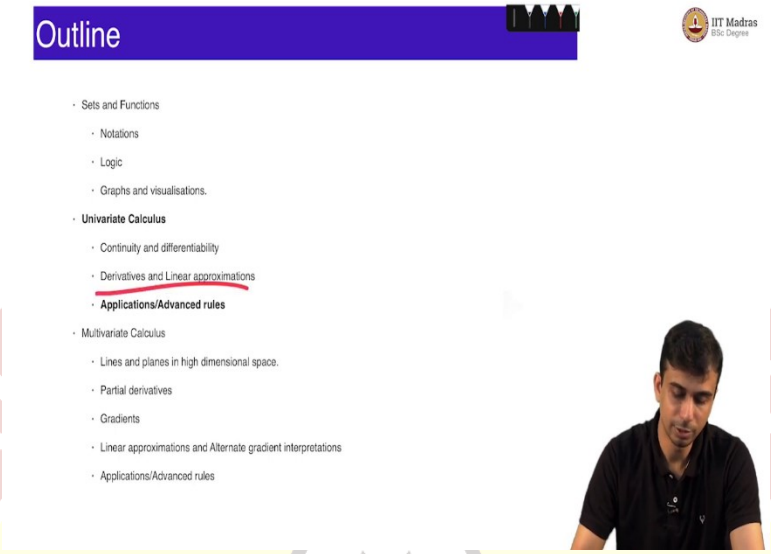






This would no longer be true, because of the same reason why, L x * of f is only an approximation of f around x equals x *. As you move farther and farther away from x *, your Lx * of f at evaluated at x is not going to be exactly= f of x, it is going to be different from f of x.

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So, with that, we have come to the end of derivatives and linear approximations. We will have a look at some slightly more advanced rules and applications of the derivative and linear approximations now.